

JOSHUA KARTHIK ASHOK

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Aspiring Backend & AI Systems Engineer passionate about building scalable applications using advanced backend architectures, cloud technologies, and SAP ABAP for enterprise-grade workflows.

Education

Amrita Vishwa Vidyapeetham

B.Tech in Computer and Engineering

- CGPA: 7.61 (Expected date of graduation: July 2027)

Aug 2023 – Present

Coimbatore, TN

Work Experience

TrustyBytes

Full Stack Developer Intern

- Engineered **React frontends** and **Node.js/MongoDB APIs**, achieving **20–30% faster UI load times** and **25% improved API reliability** through optimized components, JWT auth hardening, and structured data validation.
- Strengthened product quality by adding **70% test coverage** for new features and maintaining CI pipelines, reducing deployment issues by **40%** while contributing to Agile sprint planning and code reviews.

Hyundai

SAP ABAP Intern

- Engineered and enhanced **SAP ABAP artifacts** — including **SmartForms**, **DDIC objects**, **BAPIs**, **BADIs**, and **user-exits** — streamlining core manufacturing and procurement workflows and enabling **30% reduction in manual processing overhead** across SAP modules (MM, SD).
- Optimized legacy ABAP programs through **performance tuning**, **SQL trace analysis**, and **buffering strategies**, improving **runtime efficiency by 20–25%** and elevating system reliability in production-grade environments.

Projects

ClarityLegal

[GitHub](#)

AI-Driven Legal Document Intelligence Platform

- Engineered an AI-powered legal analysis system using **Vertex AI (Gemini)**, **OCR**, **RAG search**, **FastAPI**, and **Next.js**, enabling **<5s document processing** and structured extraction of summaries, risks, and key clauses.
- Developed **clause extraction**, **risk scoring**, and **grounded Q&A** modules with near-zero hallucination; implemented semantic search over multi-document embeddings for real-time cross-document retrieval. Validated on SaaS, NDA, and employment contracts as a commercial prototype for SMEs.

RouteX

[GitHub](#)

AI-Powered Maritime Fleet Optimization Platform

- Built an AI-driven coastal fleet optimization system using **Google OR-Tools (CP-SAT)**, **FastAPI**, **Next.js**, and **MongoDB**, solving HPCL's 9-tanker, 17-port problem with **~726 feasible route patterns** and **40–50% cost reduction** over manual planning.
- Developed a **constraint programming engine** with multi-objective support (cost, time, emissions), solver profiles (15s–300s), and parallel route generation; enforced single-port loading and max two-port discharge with **100% demand satisfaction**.
- Implemented real-time maritime visualization with **interactive route playback**, **Gantt-based fleet scheduling**, and **Celery/Redis** async processing; integrated **searoute-py** for sea distances. **Top 34 in poWer Lab 2.0 – HPCL Challenge 7.1.**

OptiWare

[Live](#) [GitHub](#)

Edge-AI Warehouse Automation

- Developed an edge-AI warehouse automation system using **Sony IMX-500**, **Raspberry Pi**, and **TensorFlow Lite**, achieving **sub-200ms inference latency** with high-precision item detection for real-time sorting.
- Integrated **PostgreSQL** and **AWS (SNS/S3/Lambda)** to enable **<1.5s event propagation**, automated alerting, and operator dashboards; optimized edge pipeline to reduce overhead by **30%**.

modernERP

[Live](#) [GitHub](#)

Cloud ERP with AI & ESG Analytics

- Architected an AI-assisted ERP for MSMEs targeting **25–40% reduction in procurement cycle time** through demand forecasting and intelligent approval routing; designed **blockchain-backed audit trails** and **ESG analytics** enabling **immutable transaction logging** and **50% reduction in compliance overhead**.

Publications

Navigating the Dual Realities of Image Generation and Deepfake Detection

[Research Paper](#)

- Conducted a comparative study of GAN- and diffusion-based image generation models, identified key robustness gaps in deepfake detection (adversarial vulnerability, compression artefacts, cross-model inconsistency), and proposed methodological directions—spanning multimodal cues and feature-level analysis—to improve forensic reliability.

Achievements & Certifications

HP poWer Lab 2.0 (HPCL) — Top 34 / 40,000+ Semi-Finalist (Project: RouteX)

[LinkedIn](#)

SEED Business Challenge 2026 (University of Maryland) — Winner

[LinkedIn](#)

Google Gen AI Exchange Hackathon — Top 100 / 30,000+ (Project: ClarityLegal)

[LinkedIn](#)

Sony AITRIOS Hackathon — Top 10 / 10,000+ (Project: OptiWare)

[LinkedIn](#)

Odoo Hackathon — Top 10 / 1,500 (Full-stack rental system)

[Google Drive](#)

SIH 2024 — Selected to Regionals (Project: modernERP)

[LinkedIn](#)

IITM Foundational Degree — Foundational in Data Science

[View Certificate](#)

Technical Skills

Languages: Python, TypeScript/JavaScript, C/C++ (familiar: Java, SAP ABAP)

Backend: FastAPI, Node.js (Express), Next.js API Routes

Databases: PostgreSQL, MongoDB, Firestore, Redis

AI/ML & CV: OpenCV, TensorFlow, TensorFlow Lite, OR-Tools (CP-SAT), OCR pipelines, searoute-py

Cloud/DevOps: Docker, Git, CI/CD, Vercel, AWS (SNS/S3/Lambda), GCP Cloud Run, Celery/Redis

Letter of Recommendation

07th April 2026

To Whom It May Concern,

I am pleased to recommend Joshua Karthik Ashok for the internship opportunity at HP Power Lab (HPCL). I have had the privilege of teaching Joshua in my Theory of Computation course, where he stood out as one of the most technically driven and intellectually curious students in the cohort.

Joshua's ability to apply complex theory to real-world systems was evident when he led development of RouteX — an AI-powered maritime fleet optimization platform — as part of Team HP at HP Power Lab 2.0. The solution leveraged constraint programming (Google OR-Tools CP-SAT), energy-aware heuristics, and parallel branch-and-bound search to tackle fleet routing at scale. The team advanced to the semifinals in Pune, reflecting both the technical rigor and practical relevance of their work. This experience makes Joshua a particularly strong fit for HPCL's applied research environment.

Beyond RouteX, Joshua brings cross-domain engineering fluency — from SAP ABAP development at Hyundai and AI/ML pipelines at TrustyBytes, to frontier work in decentralized systems using the Internet Computer Protocol. He has also earned finalist positions at the Sony AITRIOS Hackathon and the Odoo Hackathon, consistently demonstrating the ability to deliver functional prototypes under pressure.

Joshua is a proactive problem-solver and collaborative team player with a genuine passion for building impactful systems. I recommend him without reservation and am confident he will be a strong asset to your team.

Please feel free to contact me for any further information.

Yours sincerely,



Krishna Priya G

Assistant Professor

Department of Computer Science Engineering

Amrita School of Computing

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**Agriculture | Allied Health Sciences | Arts | Ayurveda | Business | Commerce | Computing | Dentistry
Engineering | Commerce | Humanities | Life Sciences | Media & Communication | Medical Sciences | Nano Sciences
Nursing | Pharmacy | Physical Sciences | Social Sciences | Spiritual Studies | Sustainable Development**